5.2(1): (a) Both onto & one-to-one

(b) Not one to one but onto

(c) No 2 cars can have same number so it is one-to-one but not onto as there are many sequences S that might not have been issued yet.

5.2(2)(a) Not one to one & not onto

(b) f is not one to one but onto. Same case for g.

(c) Both ✓

5.2(3)(b) $f(x) = x^2 - 2x$ | $f: \mathbb{R} \to \mathbb{R}$. Not one-to one as example both 2 & -2 gives same answer due to x^2.

To check onto, we must separate x on one side in equation $x^2 - 2x = a$ for arbitrary a (output). Otherwise we cannot find any preimage if x lies on both sides. We can use full square

$\to x^2 - 2x + 1 = a + 1$

$\to (x-1)^2 = a+1$

$\to x - 1 = \pm\sqrt{a+1}$

$x = 1 \pm \sqrt{a+1}$. This function is not onto as it cannot output numbers like -2. putting $a = -2$ gives $1 \pm \sqrt{-1}$ which is not real.

(c) Not onto as $f(2.1) = f(2.2) = 2$ for example. It is one to one as for every integer n, there is $n \le x < n+1$.

5.2(4)(b) f is one-to-one. This can be seen by brute force but we can prove it by contradiction too. To prove $\forall X_1 \forall X_2 [f(X_1) = f(X_2) \to X_1 = X_2]$

Supp $X_1 \in B \wedge X_2 \in B$. Note that $B = P(A)$ so

$X_1 \subseteq A \wedge X_2 \subseteq A$.

Supp $f(X_1) = f(X_2) \to A \setminus X_1 = A \setminus X_2$

$\forall x [(x \in A \wedge x \notin X_1) \leftrightarrow (x \in A \wedge x \notin X_2)]$ — ①　　$X_1 = X_2$

Supp $X_1 \neq X_2 \to \exists x [(x \in X_1 \wedge x \notin X_2) \vee (x \notin X_1 \wedge x \in X_2)]$

Case 1) supp $x \in X_1 \wedge x \notin X_2$. As $X_1 \subseteq A$ so $x \in A$. Then by $\Leftarrow$ of ①, $x \in A \wedge x \notin X_1$. And this is contradiction over assumption that $x \in X_1$.

Case 2) Similar to 1.

~~This is onto as for any $X \in B$, we have $f(X) = A \setminus X \in P(A)$. As $A \subseteq A$ & clearly $A \setminus X \subset A$ so $A \setminus X \in P(A)$.~~

Actually we have to prove $\forall Y \in B\ \exists X \in B\ (F(X) = Y)$

Let Y be arbitrary. To find X. Look at small example, we will see that $X = A \setminus Y$ works.

$F(X) = A \setminus X = A \setminus (A \setminus Y) = Y$. We can prove $A \setminus (A \setminus Y) = Y$ when $Y \subseteq A$. This is easy proof.

(c) $f : R \to (R \times R)$

$f(x) = (x+1, x-1)$

It is one to one as supp $x_1, x_2 \in R$ & $f(x_1) = f(x_2)$

then $(x_1+1, x_1-1) = (x_2+1, x_2-1)$

which means $x_1 + 1 = x_2 + 1 \to x_1 = x_2$

It is not onto as it clearly only outputs pairs of form $(x+1), (x-1)$. It cannot exhaust R^2. Example: assume it outputs $(2, 4)$ gives

$x + 1 = 2$ & $x - 1 = 4$

$x = 3$, $x = 5$ contradiction

The key to identification of these contradictions lie in trying to prove onto or playing around with function or checking edge cases or values.

Q 5.2(5) $A = R \setminus \{1\}$, $f : A \to A$, $f(x) = \frac{x+1}{x-1}$

(a) To show one to one $\forall x_1 \forall x_2 \in A\ [f(x_1) = f(x_2) \to x_1 = x_2]$

Let $x_1 \in A, x_2 \in A \wedge f(x_1) = f(x_2)$ which means

$\frac{x_1+1}{x_1-1} = \frac{x_2+1}{x_2-1} \to (x_1+1)(x_2-1) = (x_2+1)(x_1-1)$

$\to \cancel{x_1x_2} - x_1 + x_2 - \cancel{1} = \cancel{x_1x_2} - x_2 + x_1 - \cancel{1}$

$2x_1 = 2x_2 \to x_1 = x_2$

To see f is onto $\quad \forall y \in A\ \exists x \in A\ [f(x) = y]$

supp $y \in A$. To find x, we can supp output y

& then solve for $x \to \frac{x+1}{x-1} = y \to x+1 = y(x-1)$

$\to 1+y = xy - x \to 1+y = x(y-1) \to x = \frac{1+y}{y-1}$

So let $x = \frac{1+y}{y-1}$ then

$$f(x) = f\left(\frac{1+y}{y-1}\right) = \frac{\frac{1+y}{y-1}+1}{\frac{1+y}{y-1}-1} = \frac{\frac{1+y+y-1}{\cancel{y-1}}}{\frac{1+y-y+1}{\cancel{y-1}}} = \frac{2y}{2} = y$$

(b) $f \circ f = i_A$

$\to$ supp $(x,y) \in f \circ f$. $\quad \to (x,y) \in i_A$

iff $\exists z \in A\ ((x,z) \in f \wedge (z,y) \in f)$ $\quad \longrightarrow$ ~~iff $(x,y) \in i_A$~~ $x = y$

iff $f(x) = z \wedge f(z) = y$

iff $\frac{x+1}{x-1} = z$ ① $\wedge$ $\frac{z+1}{z-1} = y$ ②

As seen earlier, ① yields $x = \frac{z+1}{z-1} = y$ so $x = y$. so $(x,y) \in i_A$

$\leftarrow$ supp $(x,x) \in i_A$ ($\because$ can take x,y too & $x = y$) $\to (x,x) \in f \circ f$.

iff $\exists z \in A[(x,z) \in f \wedge (z,x) \in f]$

iff $\exists z \in A[f(x) = z \wedge f(z) = x]$

let $z = \frac{x+1}{x-1}$ then clearly $f(x) = \frac{x+1}{x-1} = z$

& $f(z) = f\left(\frac{x+1}{x-1}\right) = \frac{\frac{x+1}{x-1}+1}{\frac{x+1}{x-1}-1} = x$ (as seen earlier) so $(x,x) \in f \circ f$.

5.2(9)(a) $A = P(R)$, $B = P(A) = P(P(R))$

$f: B \to A$, $f(F) = \cup F$

(a) $f(\{\{1,2\},\{3,4\}\}) = \{1,2,3,4\} \in P(R) = A$

(b) f is not one to one because example if $F_1 = \{\{1\},\{2\}\}$

& $F_2 = \{\{1,2\}\}$ then $f(F_1) = f(F_2)$ but $F_1 \neq F_2$.

Onto

$\rightarrow \forall X \subseteq A \; \exists F \subseteq B \, (f(F) = X)$

Supp $X \subseteq A$. Let $F = \{X\}$ then $f(F) = \cup F = \cup\{X\} = X$. So f is onto.

5.2(10) $f: A \to B$, $g: B \to C$

(a) $g \circ f$ is onto $\rightarrow$ g is onto

$\forall c \in C \; \exists a \in A \, (g\circ f(a) = c)$ — ① $\qquad \forall c \in C \; \exists b \in B \, (g(b) = c)$

Let $c \in C$ be arbitrary. $\longrightarrow \exists b \ldots$

By ① $\exists a_1 \in A \, (g\circ f(a_1) = c)$

so $\exists b_1 \in B \, [(a_1, b_1) \in f \wedge (b_1, c) \in g]$ so let $b = b_1$ & clearly $g(b_1) = c$.

(b) $g \circ f$ is one to one $\longrightarrow$ f is one to one.

$\forall a_1 \in A \, \forall a_2 \in A \, [g\circ f(a_1) = g\circ f(a_2) \rightarrow a_1 = a_2]$ — ① $\qquad \forall a_1 \in A \, \forall a_2 \in A \, [f(a_1) = f(a_2) \rightarrow a_1 = a_2]$

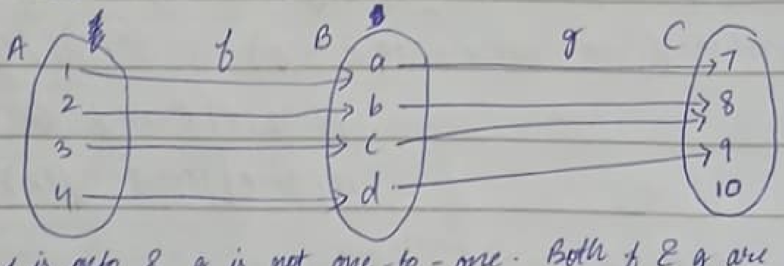
Let $a_1 \in A$, $a_2 \in A \wedge f(a_1) = f(a_2)$ $\longrightarrow a_1 = a_2$

Let $f(a_1) = f(a_2) = b \in B$

then $g\circ f(a_1) = g\circ f(a_2) = g\circ f(a_2)$ so by ①, $a_1 = a_2$

5.2(11)(a) consider contradiction

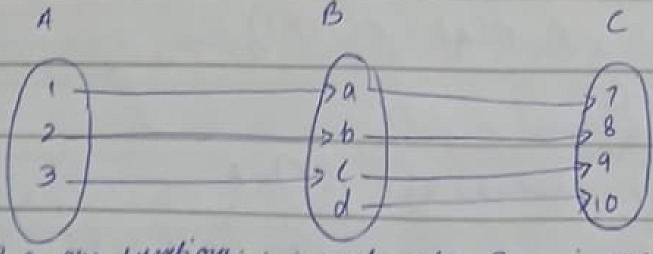
Not proof just understanding



f is onto & g is not one-to-one. Both f & g are functions

$g\circ f$ contains $(2,8)$ & $(3,8)$ so $g\circ f$ is not ~~functi~~ one to one.

(b) consider contradiction



f & g are functions. f is not onto & g is one to one. $g\circ f$ is not onto. $(x, 10) \notin g\circ f$ for any x

5.2 (11)(a) Supp f is onto & g is not one to one $\rightarrow$ $g\circ f$ is not one to one

g is not one to one means: $\exists a_1 \exists a_2 \in A [g\circ f(a_1) = g\circ f(a_2) \wedge a_1 \neq a_2]$

$\exists b_1 \in B \exists b_2 \in B [g(b_1) = g(b_2) \wedge b_1 \neq b_2]$ — ①

Since f is onto $\exists a_1 \in A (f(a_1) = b_1)$ — ②

$\exists a_2 \in A (f(a_2) = b_2)$ — ③

So we got $a_1, a_2 \in A$. Also $g\circ f(a_1) = g(f(a_1)) = g(b_1)$

Similarly $g\circ f(a_2) = g(b_2)$ but by ①, $g(b_1) = g(b_2)$ so $g\circ f(a_1) = g\circ f(a_2)$.

Finally to see $a_1 \neq a_2$. Supp $a_1 = a_2$ $\longrightarrow$ Contradiction

Then $f(a_1) = f(a_2) \Rightarrow$ so $b_1 = b_2$ which contradicts the fact that g is not one to one.

(b) f is not onto & g is one to one. $\longrightarrow$ $g\circ f$ is not onto

means $\sim \forall b \in B \exists a \in A [f(a) = b]$ $\sim \forall c \in C \exists a \in A [g\circ f(a) = c]$

or $\exists b \in B \forall a \in A [f(a) \neq b]$ — ① $\exists c \in C \forall a \in A [g\circ f(a) \neq c]$

g is one to one so $\forall b_1 \in B \forall b_2 \in B [g(b_1) = g(b_2) \rightarrow b_1 = b_2]$ — ②

from ① $b_1 \in B$ & $\forall a \in A (f(a) \neq b_1)$ — ③

Since g is fction so $\exists! c_1 \in C (g(b^1) = c_1)$ — ④ ×

~~so let $c = c_1$. Let a be arbitrary $\longrightarrow$ $g\circ f(a) \neq c_1$.~~

~~supp $g\circ f(a) = c_1$ $\longrightarrow$ contradiction~~

~~then $g\circ f(a) = g(f(a)) = g($ $\rightarrow$ No where~~

supp $g\circ f$ is onto **Proof By Contradiction**

so $\forall c \in C \exists a \in A [g\circ f(a) = c]$ — ⑤ $\longrightarrow$ Contradiction

From ④ & ⑤ $\exists a_1 \in A [g\circ f(a_1) = c_1]$ — ⑥

so $g(f(a_1)) = c_1$. Also from ④ $g(b^1) = c_1$

so $g(f(a_1)) = g(b_1)$ — ⑦

Case 1) If $f(a_1) = b_1$, this contradicts ③ $\rightarrow$ that f is not onto

Case 2) If $f(a_1) \neq b_1$, then along with ⑦, this contradicts the fact that g

is one to one i.e. it contradicts (2).

Both cases lead to contradiction when we assume that $g \circ f$ is onto. So its not.

5.2) (13) (a) f is one to one $\quad\quad$ $f|C$ is one to one.

Direct proof is difficult.

Proof by contraposition. Supp $f|C$ is not one to one. Then

$\exists c_1 \in C\ \exists c_2 \in C\ [f|C(c_1) = f|C(c_2) \wedge c_1 \neq c_2]$

Let $f|C(c_1) = b = f|C(c_2)$

then $(c_1, b) \in f \cap (C \times B)$ so $(c_1, b) \in f$. Similarly $(c_2, b) \in f$. But now

f is not one to-one. So $f|C$ is one to one.

(b) $f|C$ is onto $\quad\quad$ f is onto

$\forall b \in B\ \exists c \in C [f|C(c) = b]$ —(1) $\quad\quad$ $\forall b \in B\ \exists a \in A [f(a) = b]$

Let $b \in B$. By (1) $\exists c \in C\ (f|C(c) = b)$. So let $a = c$. Clearly $c \in A$

as $C \subseteq A$. & $f(c) = b$.

(c) Counter to (a) → let $A = \{1, 2, 3\}$, $B = \{1, 2\}$, $C = \{1, 2\}$

$f = \{(1, 2), (2, 2), (3, 2)\}$ then f is not one to one however $f|C$ is one to one.

Counter to (b) Just change $C = \{1\}$ then above, f is onto but $f|C$ is not.

Q 5.2 (18) R is equivalence on A, $g: A \to A/R$, $g(x) = [x]_R$

(a) $\quad\quad$ g is onto

Let $\quad\quad$ $\forall [a] \in A/R\ \exists x \in A (g(x) = [a])$

$[a] \in A/R$. Let $x = a$. then $g(x) = g(a) = [a]_R$.

(b) g is one to one $\longleftrightarrow$ $R = i_A$

→ Supp g is one to one.

→ Supp $(x, y) \in R$. $\longrightarrow$ $(x, y) \in i_A$

Then $x, y \in A$. Also $g(x) = [x]$ & $g(y) = [y]$ but since xRy

so $[x] = [y]$ so $g(x) = g(y)$. As g is one to one so $x = y$, so $(x, y) \in i_A$.

$\leftarrow$ $(x, y) \in i_A$ $(x, y) \in R$.

Then $x, y \in A$ & $x = y$. $g(x) = g(y)$, so $[x] = [y]$ so $x \in [y]$

so $x R y$.

$\leftarrow$ $R = i_A$ $\longrightarrow$ g is one to one

Supp $a_1 \in A$, $a_2 \in A$ & $g(a_1) = [a_1] = g(a_2) = [a_2]$. Then $a_1 R a_2$

Since $R = i_A$ so $a_1 \, i_A \, a_2$ so $a_1 = a_2$.

Note we used defs of R & equivalence classes like $xRy \Leftrightarrow x \in [y] \Leftrightarrow$

$[x] = [y]$

5.2 (23) (a) $f(n) = n$ $f: N \to N$.

f clearly maps onto n. For every $n \in N$. Same $f(n) = n$.

(b) f doesn't map onto $\mathbb{Z}$. example for $-1 \in \mathbb{Z}$, $\neg \exists n \in N (f(n) = -1)$

as $-1 \notin N$.

5.3) A proof that if $f^{-1}: B \to A$ then f^{-1} is also one to one & onto.

1- let $b_1 \in B$, $b_2 \in B$ & $f^{-1}(b_1) = f^{-1}(b_2)$ $\longrightarrow$ $b_1 = b_2$.

let $f^{-1}(b_1) = a = f^{-1}(b_2)$ then $(a, b_1) \in f$ & $(a, b_2) \in f$. As f is fcn on so

it must be that $b_1 = b_2$. so f^{-1} is one to one.

2- Supp $a \in A$. As $f: A \to B$ so $\exists! b \in B$ $(f(a) = b)$ so f^{-1} is onto.

Note that both are bijections on each other

5.3(8)(a)

$f \circ f^{-1} = i_B$

$\forall b \in B \, (f \circ f^{-1}(b) = i_B(b))$

let $b \in B$ be arbitrary

let $f^{-1}(b) = a \in A$ so $f(a) = b$ then $f \circ f^{-1}(b) = f(f^{-1}(b)) = f(a) = b = i_B(b)$

(b) Apply first half to f^{-1}. $(f^{-1} \circ f = i_A)$ $f \circ f^{-1} = i_B$

let $g = f^{-1}$ then $g^{-1} = (f^{-1})^{-1} = f$

By first half $\rightarrow$ $g^{-1} \circ g = i_B$ $(g: B \to A)$ (i_B = Domain of g)

Replacing g with f^{-1} $\rightarrow$ $(f^{-1})^{-1} \circ f^{-1} = i_B \rightarrow f \circ f^{-1} = i_B$.

5·3 (9) $g: B \to A$. $f \circ g = i_B$ $\qquad$ f is onto

supp $b \in B$ $\qquad$ $\exists a \in A\ (f(a) = b)$

As $g: B \to A$ so $\exists! a_1 \in A\ (g(b) = a_1)$

Using this, we can let $a = a_1$ or $a = g(b)$, then (A venn diagram can help recognize $g(b)$)

$f(g(b)) = f \circ g(b) = i_B(b) = b$.

Note if we have $a = b$ then we can apply any f on both sides $f(a) = f(b)$. This follows from property of functions that if $a = b$ & $f(a) \neq f(b)$ then f would not be a function.

5·3 (10) $g \circ f = i_A$, $f \circ g = i_B$. $\qquad$ $g = f^{-1}$

$\rightarrow$ let $(b, a) \in B \times A$, $(b, a) \in g$. $\qquad$ $(b, a) \in f^{-1}$

then $g(b) = a$. Applying f on both sides $\qquad$ iff $(a, b) \in f$.

$f(a) = f(g(b)) = f \circ g(b) = i_B(b) = b$. $\qquad$ iff $f(a) = b$.

so $f(a) = b$.

$\leftarrow$ let $(b, a) \in f^{-1}$ then $(a, b) \in f$ so $f(a) = b$. Applying g: $\qquad$ $(b, a) \in g$

then $g(b) = g(f(a)) = g \circ f(a) = i_A(a) = a$ $\qquad$ iff $g(b) = a$

so $g(b) = a$.

We can also use longer way by letting e.g. $f(a_1) = b$ as $f: A \to B$. Then $(a_1, a) \in g \circ f$ so $(a_1, a) \in i_A$ so $a_1 = a$. & similar for other part

5·3 (1) $R^{-1}(p)$ = The unique person q such that $(p, q) \in R^{-1}$ $(\therefore (q, p) \in R)$

" $\qquad$ " $\qquad$ p is sitting to right of q.

" $\qquad$ " $\qquad$ q is sitting to left of p

= The person to left of p.

5·3 (2) $F^{-1}(X) = Y$ such that $(Y, X) \in F$

" $\qquad X = A \setminus Y$

" $\qquad A \setminus X = A \setminus (A \setminus Y)$

" $\qquad$ " $\qquad A \setminus X = Y$ $\qquad$ so $F^{-1}(X) = A \setminus X$ also

5.3(3) $f: \mathbb{R} \to \mathbb{R}$, $f(x) = \frac{2x+5}{3}$

One to one. Supp $x_1, x_2 \in \mathbb{R}$ & $f(x_1) = f(x_2)$ $\longrightarrow x_1 = x_2$.

then $\frac{2x_1+5}{3} = \frac{2x_2+5}{3}$ so $2x_1 = 2x_2 \to x_1 = x_2$

Onto: $\frac{2x+5}{3} = y \to 2x+5 = 3y \to 2x = 3y-5$ $\forall y \in \mathbb{R}\ \exists x \in \mathbb{R}\ (f(x) = y)$

$x = \frac{3y-5}{2}$

Let y be arbitrary. Supp let $x = \frac{3y-5}{2}$. Then

$f(x) = f\left(\frac{3y-5}{2}\right) = \frac{\left(\frac{3y-5}{\cancel{2}}\right)\cancel{2}+5}{3} = \frac{3y}{3} = y$.

$f^{-1}(x) = ?$

let $f^{-1}(x) = y \to (x,y) \in f^{-1} \to (y,x) \in f \to f(y) = x \to \frac{2y+5}{3} = x$

$3x = 2y+5 \to 3x-5 = 2y \to y = \frac{3x-5}{2} = f^{-1}(x)$ (as $y = f^{-1}(x)$)

To see its really $f^{-1}(x)$.

$f \circ f^{-1}(x) = f(f^{-1}(x)) = f\left(\frac{3x-5}{2}\right) = \frac{2\left(\frac{3x-5}{2}\right)+5}{3} = x$

so $f \circ f^{-1}(x) = i_{\mathbb{R}}$ & similarly $f^{-1} \circ f = i_{\mathbb{R}}$ so f^{-1} is correct.

5.2(5) $f: \mathbb{R} \to \mathbb{R}^+$, $f(x) = 10^{2-x}$

Onto One. Let $x_1, x_2 \in \mathbb{R}$, $f(x_1) = f(x_2)$ $\longrightarrow$ $x_1 = x_2$

$10^{2-x_1} = 10^{2-x_2}$

Taking $\log_{10} \to 2-x_1 = 2-x_2$ so $x_1 = x_2$

Onto supp $y \in \mathbb{R}^+$ $\exists x \in \mathbb{R}\ (f(x) = y)$

$f(x) = 10^{2-x} = y \to 2-x = \log_{10} y \to x = 2 - \log_{10} y$

Let $x = 2 - \log_{10} y$ then $f(x) = 10^{2-2-\log_{10} y} = 10^{\log_{10} y} = y$

$\because (10^x = y \to \log_{10} y = x \to 10^{\log_{10} y} = 10^x \to$ but $10^x = y$ so $10^{\log_{10} y} = y)$

Let $f^{-1}(x) = y$ so $(y,x) \in f$ so $10^{2-y} = x$

so $2-y = \log_{10} x \to 2 - \log_{10} x = y = f^{-1}(x)$

$f \circ f^{-1}(x) = f(2 - \log_{10} x) = 10^{2-2-\log_{10} x} = x$.

Also $f^{-1} \circ f(x) = f^{-1}(10^{2-x}) = 2 - \log_{10} 10^{2-x} = 2-(2-x) = x$

so $f \circ f^{-1} = i_{\mathbb{R}^+}$ & $f^{-1} \circ f = i_{\mathbb{R}}$

5.2(6) In proving onto, $x = \frac{-2y}{3-y}$ would be obtained so y is not in range so $R = R \setminus \{3\}$.

letting $f(x) = \frac{3x}{x-2} = 3$ gives $3x = 3x - 6$ which is contradiction.

(b) set $f^{-1}(x) = y$ or $\frac{3y}{y-2} = x$ to find y.

5.2(7)(a) $f_2 \circ f_1(x) = f_2(f_1(x)) = f(x+7) = \frac{x+7}{5} = f(x)$

(b) let $f_1^{-1}(x) = y \rightarrow y + 7 = x \rightarrow y = x - 7 = f_1^{-1}(x)$

let $f_2^{-1}(x) = y \rightarrow \frac{y}{5} = x \rightarrow y = 5x = f_2^{-1}(x)$

$f_1^{-1} \circ f_2^{-1}(x) = f_1^{-1}(5x) = 5x - 7 = f^{-1}(x)$.

5.2(11) $f : A \rightarrow B$, $g : B \rightarrow A$.

(a) f is one to one $\wedge$ $f \circ g = i_B$ $\rightarrow$ $g = f^{-1}$

By 5.3.3(2), f is onto so $f^{-1} : B \rightarrow A$

$\rightarrow$ supp $(b, a) \in g$. $(b,a) \in f^{-1} \Leftrightarrow (a,b) \in f$.

$f^{-1} : B \rightarrow A$ $\exists! a_1 \in A$ $(b, a_1) \in f^{-1}$ so $(a_1, b) \in f$.

$f : A \rightarrow B$ $\exists! b_1 \in B$ $(a, b_1) \in f$ — (2)

By def of composition $\rightarrow$ $(b, b_1) \in f \circ g = i_B$ so $b = b_1$

so from (2) $\rightarrow (a, b) \in f$. $\leftarrow$ side is similar

(b) f is onto, $g \circ f = i_A$ $g = f^{-1}$

By 5.3.3(1), f is one to one.

so $f^{-1} : B \rightarrow A$.

supp $(b, a) \in g$ $(b,a) \in f^{-1} \Leftrightarrow (a,b) \in f$.

$f : A \rightarrow B$ so $(a, b_1) \in f$. so $(b_1, a) \in f^{-1}$ — (1)

but $g : B \rightarrow A$ so as $b_1 \in B$ so $\exists! a_1 \in A$ $(b_1, a_1) \in g$ (2)

By def of composition $(a, a_1) \in g \circ f = i_A$ so $a_1 = a$

so from (2) $(b_1, a) \in g$ but as $(b, a) \in g$ as well.

& g is fction so $b_1 = b$ so from (1) $\rightarrow (a, b_1) = (a, b) \in f$

$\Leftarrow$ supp $(b,a) \in f^{-1}$ $\qquad$ $(b,a) \in g$

$(a,b) \in f$ — ①

Also $g: B \to A$ so $\exists! a_1 \in A$ $(b, a_1) \in g$ — ②

But $f: A \to B$ so $\exists! b_1 \in B$ $(a_1, b_1) \in f$ — ③ → Not needed

By ① & ② $(a, a_1) \in g \circ f = i_A$ so $a = a_1$ so $(a, a_1) \in g$ by ②

(c) $f \circ g = i_B$ $\quad$ $g \circ f \neq i_A$

① First note by 5.3.3(2), f is onto $\qquad$ f is onto

Proved

② Proof by contradiction $\qquad$ f is not one to one.

Supp f is one to one

Let $(a,b) \in f$. then as $g: B \to A$ ~~so~~ $(b, a) \in g$ for $\exists! a_1 \in A$

but as $a_1 \in A$ so $\exists! b_1 \in B$ $(a_1, b_1) \in f$.

then by def of composition, $(b, b_1) \in f \circ g = i_B$ so $b = b_1$

so $(a_1, b_1) = (a_1, b) \in f$.

But f is one to one so $a = a_1$ b

As $g \circ f \neq i_A$. There are 2 cases.

Case 1) $\exists (a_1, a_2) \in A \times A \, [(a_1, a_2) \in g \circ f \wedge (a_1, a_2) \notin i_A]$

then by def of $g \circ f$ → $\exists b \, [(a_1, b) \in f \wedge (b, a_2) \in g]$

As $f: A \to B$ so $\exists! b_2 \in B$ $(a_2, b_2) \in f$.

then by def of $f \circ g$, $(b, b_2) \in f \circ g = i_B$ so $b_2 = b$.

so $(a_2, b_2) = (a_2, b) \in f$. But then under assumption that f is one to one → $a_1 = a_2$. This contradicts the fact that $(a_1, a_2) \notin i_A$.

Case 2) $\exists (a_1, a_2) \in A \times A \, [(a_1, a_2) \in i_A \wedge (a_1, a_2) \notin g \circ f]$

so $a_1 = a_2$ let $a_1 = a_2 = a$ so $(a, a) \in i_A$ $\wedge$ $(a, a) \notin g \circ f$

As $f: A \to B$ $\exists! b \in B$ $(a, b) \in f$

As $g: B \to A$ $\exists! a_1 \in A$ $(b, a_1) \in g$

As $f: A \to B$ $(a_1, b_2) \in f$.

But then $(b, b_2) \in f \circ g = i_B$ so $b_1 = b_2$

But then $(a_1, b_2) = (a_1, b) \in f$. & as by assumption f is one to one

so $a = a_1$ so $(b, a_1) = (b, a) \in g$.

But also with $(a, b) \in f$ $(a, a) \in g \circ f$ is contradiction

(3)

g is one to one

Applying part (1) of 5.3.3 to g, g is one to one.

(4) Proof by contradiction

g is not onto.

Supp g is onto so $\forall a \in A\, \exists b \in B\, (g(b) = a)$ — (1)

similar to part 2.

Case (1) $\exists (a_1, a_2) \in g \circ f \wedge (a_1, a_2) \notin i_A$.

By (1) $(b_1, a_1) \in g$ — (2)

$(b_2, a_2) \in g$.

Also by def of $g \circ f \to \exists b$ $(a_1, b) \in f \wedge (b, a_2) \in g$. (3)

then $(b_1, b) \in f \circ g = i_B$ so $b_1 = b$ But this would contradict g

being a function ((2) & (3)) so $a_1 = a_2$. This contradicts $(a_1, a_2) \notin i_A$

Case (2) $(a, a) \in i_A \wedge (a, a) \notin g \circ f$.

By (1) $\exists b (b, a) \in g$

$f: A \to B$ so $\exists! b_1 \in B$ $(a, b_1) \in f$. — (2)

$g: B \to A$ so $\exists! a_1 \in A$ $(b_1, a_1) \in g$. — (3)

so $(b, b_1) \in f \circ g = i_B$ so $b = b_1$ so $(a, b) \in f$

But then $(a, a) \in g \circ f$ which is contradiction

5.4 (11) $f: A \times A \to A$, $B \subseteq A$ $\to \exists D$ (D is closure of B)

Let $F = \{ C \subseteq A \mid B \subseteq C \wedge C$ is closed under $f \}$

Let $D = \cap F$

1) $B \subseteq D$

Let x be arbitrary. Supp $x \in B$ $\rightarrow$ $x \in \cap F \leftrightarrow \forall E \in F (x \in E)$

Supp $E \in F$. Then by def of F, $B \subseteq E$ so $x \in E$. $x \in E$

2) $\forall x \in D \forall y \in D (f(x,y) \in D)$

supp $x \in D, y \in D$ $f(x,y) \in D = \cap F$

$\leftrightarrow \forall X \in F (x \in X), \forall Y \in F (y \in Y)$ iff $\forall E \in F (f(x,y) \in E)$

Supp $E \in F$ $f(x,y) \in E$

so $x \in E$. also $y \in E$ But as $E \in F$ so E is closed under F. so any $x, y \in E$ means $f(x,y) \in E$ as well.

3) supp $I \subseteq A$, $B \subseteq I$, I is closed under F $\rightarrow D = \cap F \subseteq I$.

supp $x \in \cap F$ $\emptyset$ $x \in F$

$\forall X \in F (x \in X)$

But by def of F, $I \in F$ so $x \in I$. (I is arbitrary element of F)

5.4 (16) (a) $\forall X \subseteq N \exists Y \in E \exists Z \in I (Y \cap Z = X)$

supp if $X = \{1\}$ then we can have any infinite $Y = X \cup$ infinite set.

supp $Y = \{1, 3, 5, \ldots\}$ then let $Z = X \cup (N \setminus Y) = \{1, 2, 4, 6 \ldots\}$

so now $Y \cap Z = \{1\} = X$. Also note that 1 is forced to be element of any of Y or Z. It cannot be that 1 is not in Y & not in Z.

so let $X \subseteq N$ let $Y = X \cup (N \setminus Z)$, let $Z = X \cup (N \setminus Y)$

then $Y \cap Z = X$

$\leftarrow x \in X$ $x \in Y \cap Z$ $= x \in Y \wedge x \in Z$

clearly $x \in Y = X \cup (N \setminus Z)$. similar for Z.

$\rightarrow x \in Y \cap Z \rightarrow x \in Y \wedge x \in Z$. (4 cases) $x \in X$

Case 1) $x \in X$ (as $x \in X$)

Case 1.1) $x \in X \rightarrow$ Goal follow

Case 1.2) $x \in N \setminus Y \rightarrow$ Goal follow

Case 2) $x \in N \setminus Z$

Case 2.1) $x \in X \rightarrow$ Goal follow

Case 2.2) $x \in N \setminus Y \rightarrow$ Then $x \in N \wedge x \notin Z \wedge x \notin Y$ but this is not possible as $x \in Y \cap Z$ so $x \in Y \wedge x \in Z$.

(b) closure of I under $\cap$?

As in part (a), we saw for any arbitrary $X \subseteq N$, there are Y & Z in I so X has to be included in I to make it closed so I must contain every $X \subseteq N$ to be closed so closure of $I = \{X \subseteq N \mid X$ can be finite, infinite or $\phi\} = P(N)$

To prove $I \subseteq P(N)$ & that $P(N)$ is closed is easy as it will follow from (a).

Supp $G \subset P(N)$, $I \subseteq G$ & G is closed under $\cap$ $\rightarrow P(N) \subseteq G$

~~As I~~ supp $X \in P(N)$ as $I \subseteq P(N)$ so $X \in I$ & as $I \subseteq G$ so $X \in G$.

5.4(1) $f: R \rightarrow R$ $\quad f(x) = \frac{x+1}{2}$

(a) $Z \rightarrow$ No example $2 \in Z$ but $\frac{2+1}{2} \notin Z$.

(b) $Q \rightarrow$ Yes for any $x = \frac{p}{q}$, $\frac{\frac{p}{q}+1}{2} = \frac{p+q}{2q} \in Q$

(c) $\{x \in R \mid 0 \leq x < 4\} \rightarrow$ Yes, for any $0 \leq x < 4$

$1 \leq x+1 < 5 \rightarrow 0.5 \leq \frac{x+1}{2} < 2.5$

(d) $\{x \in R \mid 2 \leq x \leq 4\} \rightarrow$ No, e.g. for $x = 2$, $\frac{2+1}{2} = 1.5 \ngeq 2$.

5.4) 2) $f: P(N) \rightarrow P(N)$, $f(X) = X \cup \{17\}$

(a) Yes (b) Yes (c) No (d) Yes

5.4) 3) $\{-1, 0, 1, 2\}$.

5.4) 4) By exercise 12 of section 4.3, all 3 are closed.

5.4) 5) Yes, ϕ is closed under f trivially (as on pg 73)

5.4) 12) $F = \{$functions from A to $A\}$, $C \subseteq A$

C is closed under F if $\forall f \in F \, \forall x \in C \, (f(x) \in C)$

(a) $f(x) = x+1$, $g(x) = x-1$

Closure of $\{0\}$ under f & g $= Z$ (as +ve & -ve numbers have to be added)

(b) $\forall n \in N \, [f_n : P(N) \rightarrow P(N) \mid f_n(X) = X \cup \{n\}]$

$F = \{f_n \mid n \in N\} = \{f_0, f_1 \cdots \}$

Closure of $\{\emptyset\}$ under F

$f_0(\emptyset) = \emptyset \cup \{0\} = \{0\}$. $f_1(\emptyset) = \{1\}$ . . . . .

Also Now $f_2(\{1\}) = \{1, 2\}$ . . .

so closure is $P(N)$ as every subset can be made

5.4) 13) $B \subseteq A$

(a) let $\mathcal{G} = \{C \subseteq A \mid B \subseteq C \wedge \forall f \in F \, \forall x \in C \, (f(x) \in C)\}$.

we have to prove ① $B \subseteq \cap \mathcal{G}$

② $\cap \mathcal{G}$ is closed under F

③ $\forall D \subseteq A \, [\, (B \subseteq D \wedge D \text{ is closed under } F) \to \cap \mathcal{G} \subseteq D \,]$

1) supp $x \in B$. $\longrightarrow$ $x \in \cap \mathcal{G}$

supp $X \in \mathcal{G}$ then $B \subseteq X$ so $x \in X$ $\quad \forall X \in \mathcal{G} \, (x \in X)$

2) $f \in F$, $x \in \cap \mathcal{G}$ $\longrightarrow$ $f(x) \in \cap \mathcal{G}$

$\forall X \in \mathcal{G} \, (x \in X)$ $\quad \forall Y \in \mathcal{G} \, (f(x) \in Y)$

$Y \in \mathcal{G}$. so Y is closed under F. Also $x \in Y$ $\quad f(x) \in Y$

so $f(x) \in Y$.

3) $B \subseteq D$, D is closed. $\longrightarrow$ $\cap \mathcal{G} \subseteq D$

$\forall X \in \mathcal{G} \, (x \in X)$ $\quad x \in D$

then as $D \in \mathcal{G}$ so $x \in D$ (by def of $\mathcal{G}$).

(b) C_f = closure of B under f. $\quad \bigcup_{f \in F} C_f \subseteq C$

supp $x \in \bigcup_{f \in F} C_f$ $\quad x \in C$

~~$\exists X \in \bigcup_{f \in F} C_f \, (x \in X)$~~

$\exists f \in F \, (x \in C_f) \to f_0 \in F \, (x \in C_{f_0}) \to f_0(x) \in C_{f_0}$ (as C_{f_0} is closed under f_0)

since C is closed under F & $f_0 \in F$ so $\forall y \in C \, (f_0(y) \in C)$

If we look at any cases like $x \in B \vee x \notin B$ or proof by contradiction like $x \notin C$. it leads nowhere. Carefully looking at definitions of closures, we can understand we have to use last property of closure

As clearly $C \subseteq C$, Also $B \subseteq C$ & C is closed under every function so obviously its closed under f_0. Then since C_{f_0} is smallest subset of A (closure) so $C_{f_0} \subseteq C$ so $x \in C$.

Note we just proved that $\forall f \in F (C_f \subseteq C)$.

(c)

$\cancel{\ }$ $\forall f \in F \, \forall x \in \bigcup_{f \in F} C_f \, [f(x) \in \bigcup_{f \in F} C_f]$

Supp $f \in F$. Supp $x \in \bigcup_{f \in F} C_f$. $\rightarrow f(x) \in \bigcup_{f \in F} C_f$

then $\exists f' \in F (x \in C_{f'})$ $\exists f \in F (f(x) \in C_f)$

Since $C_{f'}$ is closed under f' so $f(x) \in C_{f'}$. so just let $f = f'$.

(d)

wrong proof as $x \in C_{f'}$ means $f'(x) \in C_{f'}$) so this doesnt proof for arbitrary f. We canot say $f(x) \in C_f$ for arbitrary f from $f'(x) \in C_f$

Counterexample to part (c) & (d)

$A = \{1, 2, 3\}$, $f = \{(1,1), (2,3), (3,3)\}$, $g = \{(1,2), (2,2), (3,3)\}$.

$F = \{f, g\}$, $B = \{1\}$

then $C_f = \{1\}$, $C_g = \{1, 2\}$, $C = \{1, 2, 3\}$.

$\bigcup_{f \in F} C_f = \{1, 2\} \neq C$

5.4(14) $\mathbb{Z}$ $(\because f(0,+1) = -1, f(0,2) = -2 \cdots)$

5.4(15) $\mathbb{Q}^+$ $(\because$ for $x = p, y = q$, $\frac{x}{y} = \frac{p}{q} \in \mathbb{Q}^+$ for $(x, y \in \mathbb{Z}^+)$

5.4(19) Reflexive Yes $\rightarrow$ Any $(a,a) \in R$ & $(a,a) \in S$ then $(a,a) \in S \circ R$.

Symmetric No $\rightarrow$ e.g. $R = \{(a,b), (b,a)\}$ & $S = \{(b,c), (c,b)\}$

then $S \circ R = \{(a,b)\}$ is not symmetric

Transitive. No e.g. $R = \{(a,b), (c,c)\}$, $S = \{(a,a), (b,c)\}$

Both are transitive but $R \circ S = \{(a,b), (b,c)\}$ is not as $(a,c) \notin R \circ S$

5.4) 21) (a) F: set of functions from $A \times A \rightarrow A$

$C \subseteq A$

let $G = \{C \subseteq A \mid B \subseteq C \wedge \forall f \in F \, \forall x \in C \, \forall y \in C \, (f(x,y) \in C)\}$

As $A \in G$ so $G \neq \phi$

1) closure is $\cap G$ — $B \subseteq \cap G$

supp $x \in B$ — $\forall X \in G (x \in X)$

supp $X \in G$ — $x \in X$

then $B \subseteq G$ so $x \in G$.

2) — $\cap G$ is closed under F

supp $f \in F$, $x \in \cap G$ & $y \in \cap G$ — $f(x,y) \in \cap G$

$\forall X \in G (x \in X)$, $\forall Y \in G (y \in Y)$ — $\forall Z \in G (f(x,y) \in Z)$

supp $Z \in G$ then $x \in Z$ & $y \in Z$ — $f(x,y) \in Z$

Also as $Z \in G$ so Z is closed under F, so $f(x,y) \in Z$.

3) similar to previous exercises.

(b) 1) $Q \cup \{\sqrt{2}\} \subseteq \{a + b\sqrt{2} \mid a, b \in Q\}$. — let this be $Q(\sqrt{2})$ as in book

supp $x \in Q \cup \{\sqrt{2}\}$

Case 1) $x \in Q$ then $x = x + 0\sqrt{2} \in Q(\sqrt{2})$ as $0 \in Q$

Case 2) $x \in \{\sqrt{2}\}$ then $x = \sqrt{2} = 0 + 1\sqrt{2} \in Q(\sqrt{2})$

2) $\forall h \in F \; \forall x \in Q(\sqrt{2}) \; \forall y \in Q(\sqrt{2}) \; [h(x,y) \in Q(\sqrt{2})]$.

supp $h \in F = \{f, g\}$

Case 1) $h = f$. → existential instantiation

supp $x, y \in Q(\sqrt{2})$ then $x = a_1 + b_1\sqrt{2}$ & $y = a_2 + b_2\sqrt{2}$

so $x + y = (a_1 + a_2) + (b_1 + b_2)\sqrt{2} \in Q(\sqrt{2})$

Case 2) $h = g$

Then supp as above. $xy = a_1 a_2 + a_1 b_2\sqrt{2} + a_2 b_1\sqrt{2} + b_1 b_2(2)$

$= (a_1 a_2 + 2 b_1 b_2) + (a_1 b_2 + a_2 b_1)\sqrt{2} \in Q(\sqrt{2})$

3) supp $Q \cup \{\sqrt{2}\} \subseteq D \subseteq R$ & D is closed under F (so D is closed under both f & g)

$\longrightarrow Q(\sqrt{2}) \subseteq D$

supp $x \in Q(\sqrt{2})$ then $x = a + b\sqrt{2}$ for $a, b \in Q$ — $x \in D$

Note that $a, b \in Q$ so $a, b \in D$. Also $Q \cup \{\sqrt{2}\} \subseteq D$ so $\sqrt{2} \in D$

Since D is closed under g so $b\sqrt{2} \in D$ (as $b \in D$ & $\sqrt{2} \in D$)

Since D is closed under f so $\overset{=x}{a + b\sqrt{2}} \in D$ as ($a \in D$ & $b\sqrt{2} \in D$)

Hence Proved!

5.5) 1) $f: A \to B$, $W \subseteq A$, $X \subseteq A$

(a) $f(W \cup X) = f(W) \cup f(X)$

→ supp $b \in f(W \cup X)$ then $b \in f(W) \vee b \in f(X)$

$\exists x \in W \cup X\,(b = f(x)) = \exists x[(x \in W \vee x \in X) \wedge f(x) = b]$ $\exists w \in W(f(w) = b) \vee \exists x \in X(f(x) = b)$

case 1) supp $x \in W \wedge f(x) = b$ then $b \in f(W)$

case 2 is similar

← supp $b \in f(W) \vee b \in f(X)$ $b \in f(W \cup X)$

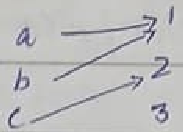
case 1) $b \in f(W)$ then $\exists w \in W(f(w) = b)$ so $w \in W \cup X$ (as $w \in W$) so $b \in f(W \cup X)$

case 2 is similar

(b) No. in → side, we have $b \in f(W \setminus X)$ then $\exists x_1 \in W \setminus X\,(f(x_1) = b)$

but we have to prove $\forall x \in X\,(f(x) \neq b)$ for b to be in $f(W) \setminus f(X)$

($\because b \notin f(X)$)

Counterexample:  $W = \{a, b\}$ $f(W) = \{1\}$ $X = \{b, c\}$ $f(X) = \{1, 2\}$

→ side would be true if f was one to one. (A proof by contradiction can be provided)

(c) $W \subseteq X \leftrightarrow f(W) \subseteq f(X)$

→ side is easy. Any $w \in W$ also implies $w \in X$.

← side fails due to f being not one to one. If it is one to one, the theorem will be correct. Let's see

$f(W) \subseteq f(X)$ — ① $W \subseteq X$ $w \in X$

supp $w \in W$. as $W \subseteq A$ so $w \in A$

$f: A \to B$ so $\exists! b_1 \in B(f(w) = b_1)$ so $b_1 \in f(W)$

by ①, $b_1 \in f(X)$ so $\exists x \in X(f(x) = b_1)$. We cannot prove $W = X$. It is only true if f is one to one then $f(w) = b_1 = f(x)$ so $x = w$. Counterexample is similar to (b)

5.5) 2) $f: A \to B$, $Y \subseteq A$, $Z \subseteq B$

(a) $$f^{-1}(Y \cap Z) = f^{-1}(Y) \cap f^{-1}(Z)$$

$\to$ side is similar to proof of 5.5 (1) (a).

$\leftarrow$ supp $a \in f^{-1}(Y) \wedge a \in f^{-1}(Z)$ $\qquad$ $a \in f^{-1}(Y \cap Z)$.

then $\exists y \in Y (f(a) = y)$ $\exists z \in Z (f(a) = z)$

but f is fction so $y = z$. so $y = z \in Y \cap Z \wedge f(a) = y = z$ so $a \in f^{-1}(Y \cap Z)$

(d) $$Y \subseteq Z \leftrightarrow f^{-1}(Y) \subseteq f^{-1}(Z)$$

$\to$ supp $a \in f^{-1}(Y)$ then $\exists y \in Y (f(a) = y)$. As $Y \subseteq Z$ so $y \in Z$. so $a \in f^{-1}(Z)$

$\leftarrow$ This side fails.

supp $y \in Y \subseteq B$ $\qquad \longrightarrow \qquad$ $y \in Z$

f^{-1} may not be fction so $f^{-1}(y)$ may not be defined. This side is true iff ~~f^{-1} is fuction~~ That is) f is surjective (onto). Then we are done

$\exists a_1 \in A$ $f^{-1}(y) = a_1$ or $f(a_1) = y$. (also to conclude $y \in Z$, injec)

Then $a_1 \in f^{-1}(Y)$ so $a_1 \in f^{-1}(Z)$ so $\exists z \in Z$ $(f(a_1) = z)$

so $y = f(a_1) = z \in Z$.

f need not be one to one

Counterexample if f is not onto.

$1 \to a$ $\qquad$ choose $Y = \{c\}$

$2 \to a$ $\qquad$ & $Z = \{a\}$

$3 \to b$

c

5.5) 3) $f^{-1}(f(X)) = X$?

supp $a \to 1$, $b \to 1$ then $f^{-1}f(\{a\}) = f^{-1}(\{1\}) = \{a, b\} \neq a$.

so f need to be one to one.

supp if f is one to one & supp $x \in f^{-1}(f(X))$ $\qquad$ $x \in X$

let $f(X) = Y$. so $x \in f^{-1}(Y)$ so $\exists y \in Y$ $(x, y) \in f$. (or $f(x) = y$)

as $y \in Y = f(X)$ so $\exists x_1 \in X$ $(x_1, y) \in f$. but f is one to one so $x_1 = x \in X$.

$\leftarrow$ supp $x \in X$ $\qquad$ $x \in f^{-1}(f(X))$

since $f: A \to B$ & $x \in A$ as well $(X \subseteq A)$ $\exists y \in f(X)\ (f(x) = y)$

so $\exists! y \in B\ (f(x) = y)$ $\exists x \in X\ (f(x) = y)$

As $x \in X$ & $f(x) = y$ so $y \in f(X)$ & clearly as $f(x) = y$ so $x \in f^{-1}(\underbrace{f(X)}_{Y})$

5.5) 4) $f(f^{-1}(Y)) = Y$?

No. example $\begin{matrix} 1 \to a \\ 2 \to b \\ c \end{matrix}$ if $Y = \{c\}$ then $f(f^{-1}(\{c\})) = f(\phi) = \phi \neq Y$

it is if Y is onto.

5.5) 5) C is closed under f $(\forall x \in C\, f(x) \in C)$ $\longrightarrow$ $f(C) \subseteq C$

$\to$ supp $x \in f(C)$ $\exists y \in C\ (f(y) = x)$. $\longrightarrow$ $x \in C$

since $y \in C$ but C is closed so $f(y) = x \in C$.

~~$\leftarrow$ supp $x \in C$~~

$x \in f(C)$

~~$\exists y \in C\ (f(y) = x)$~~

$f(C) \subseteq C$ $\longrightarrow$ $C \subseteq f^{-1}(C)$

supp $x \in C \subseteq A$ so $x \in A$ as well $x \in f^{-1}(C)$

C is closed so $f(x) \in C$ so. $x \in A \wedge \exists y \in C\ (f(x) = y)$

let $y = f(x) \in C$

$C \subseteq f^{-1}(C)$ -1 $\longrightarrow$ C is closed.

$f(x) \in C$

supp $x \in C$

1 $\to$ $x \in f^{-1}(C) \to \exists y \in C\ (f(x) = y)$

so $f(x) = y \in C$.

5.5) 7) $f^{-1}(Y) = \{a \in A \mid \exists y \in Y\ (f(a) = y)\}$ (inverse image under f) -1

To treat image under f^{-1}. supp $g: B \to A$. & let $g = f^{-1}$

then $g(Y) = \{a \in A \mid \exists y \in Y\ (g(y) = a)\}$ (forward image of g.)

$= f^{-1}(Y) = \{a \in A \mid \exists y \in Y\ (f^{-1}(y) = a)\}$.

Forward image $= \{a \in A \mid \exists y \in Y\ (f(a) = y)\} = f^{-1}(Y)$ (inverse image from 1)

as $f^{-1}: B \to A$ so $f^{-1}(y) = a$ iff $f(a) = y$